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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

March 2023 - Paper 2 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-FS-Maths-L2-Paper-2-March-2023

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Section A

Answer **all** questions in the spaces provided.

1 Circle the **largest** number.

[1 mark]

2.4 2.41 2.396 2.409

Q1 Worked Solution:

Two million three hundred (and)
seven thousand (and) forty nine
[B1] [B1]

2 Write the number two million, four hundred and eight thousand in digits.

[1 mark]

Q2 Worked Solution:

Alternative method 1
0.614 with 0.121 and 0.114 seen
0.614 with 0.121 and 0.621seen
0.614 with 0.114 and 0.386 seen
[B1] [B2]

Answer _____

3 Write the ratio 150 : 240 in its simplest form.

[2 marks]

Q3 Worked Solution:

0.379 + 0.614
or 0.4965
0.614 with 0.4965 seen
[M1] [A1]

Answer _____ : _____

Turn over ►

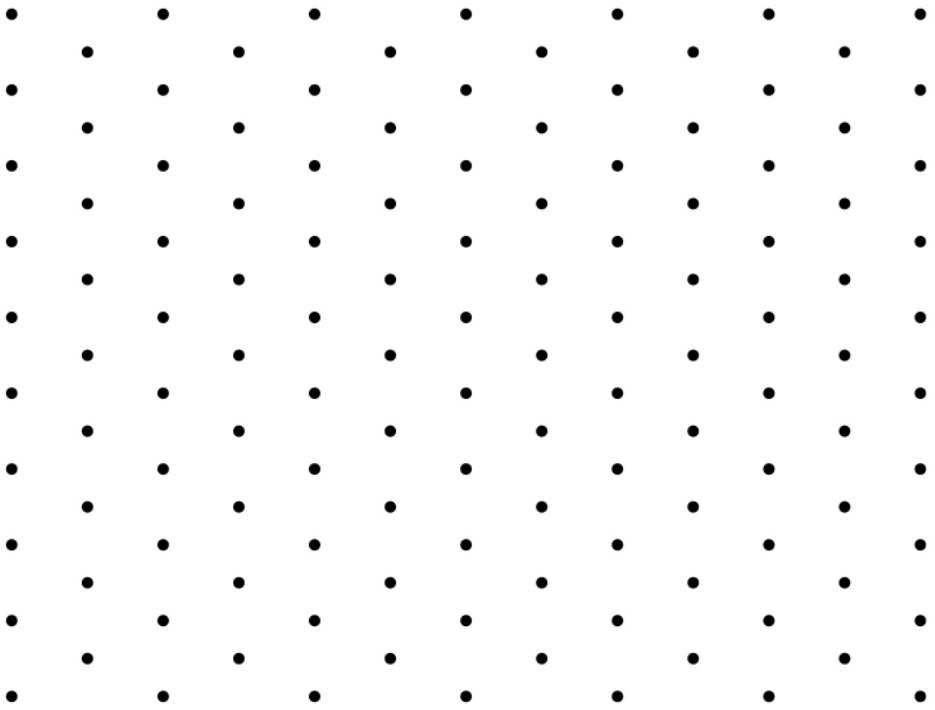
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4 A cube has side length 3 cm

Use the centimetre isometric paper to draw the cube.

[2 marks]

Q4 Worked Solution:
cm2
[B1]



5 Complete the table to show 3% as a decimal and a fraction.

[2 marks]

Percentage	Decimal	Fraction
3%		

Do not write outside the box

6 An object has mass 59.5 grams and volume 17 cm^3

Work out the density of the object.

[2 marks]

Q6 Worked Solution:

Alternative method 1

÷ +

29 (7 1) or 3(...)

29 – their 3 or 26

their 26 4.99 or 129.74

× +

29 4.25 3.90 or 127.15

(£)129.74 and (£)127.15 and Bags

and stuff

[M1] [M1] [M1] [A1]

Answer _____ grams/cm^3

7 Calculate $\frac{7}{8} + \frac{1}{4}$

Give your answer as a mixed number.

[2 marks]

Q7(a) Worked Solution:

(£)129.74 and (£)127.15 and Bags

and stuff

Alternative method 1

π 3

× 72 × ×

25

[2307, 2309.4]

× (÷

40 their [2307, 2309.4] 1000)

Answer _____

or [92 280, 92 376] or [92. 2, 92 .4]

× (×

13 6 1000) or 78 or 78 000

[A1] [M1] [M2] [A1] [M1] [M1]

12

Turn over ►

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Section B

Answer **all** questions in the spaces provided.

8 Running

Israa has started running to get fit.

8 (a) One day, Israa ran and walked for 55 minutes in the ratio

minutes running : minutes walking = 7 : 4

The table shows information about calories.

Q8(a) Worked Solution:
9 (am) and 26.6...
9 (am) and 26.7
[A1]

	Calories burned per minute
Running	16.2
Walking	5.8

Work out the total number of calories Israa burned in the 55 minutes.

[5 marks]

Answer _____ calories

Do not write outside the box

8 (b) Israa wants to buy new running shoes.

The price of the shoes is reduced in a sale.

Sale

30% off



Sale price £84

By how much, in pounds, is the price reduced?

[4 marks]

Answer £ _____

Question 8 continues on the next page

Do not write outside the box

8 (c) 180 runners take part in a race.

Each runner completes either 3 km or 5 km

The two-way table shows information about the runners.

	Child	**Adult**	**Total**
3 km	46	24	70
5 km	13		
Total	59		180

A runner is chosen at random.

Israa says,

"The probability that the runner is an adult in the 5 km race is **less than** 0.55"

Is she correct?

You **must** show your working.

[4 marks]

13

Do not write outside the box

9 Buying a house

Jack is renting a flat while he saves to buy a house.

9 (a) Jack rents a flat in town.

The table shows information about the rent of 20 other flats in the same town.

Q9(a) Worked Solution:

Rent per month (£)	Midpoint	Frequency	
Over 500 up to 550		5	
Over 550 up to 600		6	
Over 600 up to 650		7	
Over 650 up to 700		2	
		Total = 20	

45.5(0)

68 + 324 12 or 95
3090 12 or 257.5(0)
their 45.5(0) + their 95 + their

Jack pays rent of £637 per month.

How much **more** than the estimated mean of these 20 flats does Jack pay per month?

[5 marks]

257.5(0) or 398
257.5(0) – 25 – 39 – 68 – 235 or 31
398 and 367 and No
31 and No
or 45.5(0) 12) or 546
× + +
68 12 324 or 816 324 or
1140
[M1] [M1] [A1] [M1] [A1] [M1]

Answer £ _____

Question 9 continues on the next page

Do not write outside the box

9 (b) Jack plans to save a total of £18 000

He

has already saved £13 200

will save the rest from his monthly salary of £1600

will save the same amount each month for 24 months.

What **percentage** of his monthly pay does he need to save?

[4 marks]

Answer _____ %

Do not write outside the box

9 (c) Jack takes a second job.

He works each Saturday

from 9 am to 12 noon

and

from 1 pm to 5 pm

He earns £13.50 per hour.

$\frac{1}{5}$ of his pay is deducted for income tax.

How much does Jack earn each Saturday after income tax has been deducted?

[5 marks]

Answer £ _____

14

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10 **Goats**

Kira keeps goats.

10 (a) Kira needs a fenced area in her field for some of her goats.

The fenced area will be

a **square** that covers 81 m^2

in the north west corner of the field.

The centimetre square grid shows a scale drawing of Kira's field.

Q10(a) Worked Solution:
58.(...) and 60 and Yes
14.(..) and 16 and Yes
[45.9, 46] and 44 and Yes
[Method & Accuracy]

Scale: 2 centimetres represents 3 metres

N



Draw the plan of the fenced area.

[4 marks]

Do not write outside the box

10 (b) Kira has 48 goats that produce milk.

On average, each goat produces 3.21 litres of milk per day.

Using **approximations**,

estimate how many litres of milk, in total, the goats produce each month.

You **must** show your working.

[3 marks]

Answer _____ litres

Question 10 continues on the next page

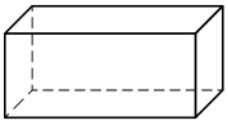
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10 (c) Kira uses some of the milk from her goats to make soap.

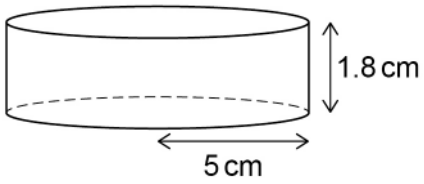
The soap is made into two different solids.

To package the soaps, Kira needs to work out the surface areas.

Soap **A**



Soap **B**

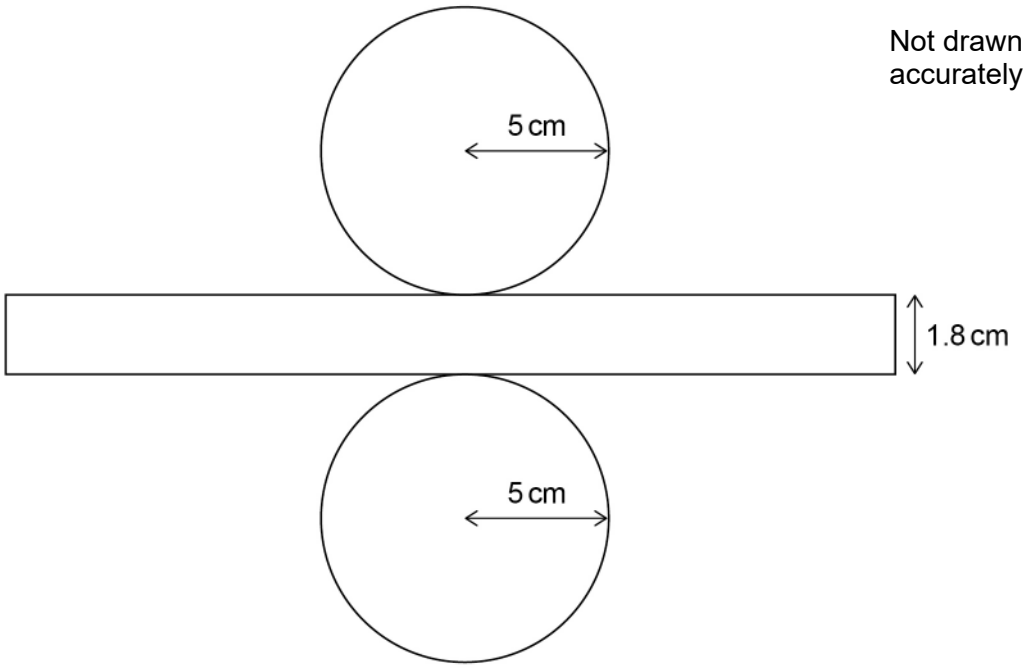


Soap A is a cuboid with a surface area of 153 cm^2

Soap B is a cylinder with a radius of 5 cm and a height of 1.8 cm

Here is a sketch of the net of the cylinder.

Not drawn accurately



Do not write outside the box

Kira says,

"The surface area of soap A is **less than** $\frac{3}{4}$ of the surface area of soap B."

Is Kira correct?

You **must** show your working.

[6 marks]

13

Turn over for the next question

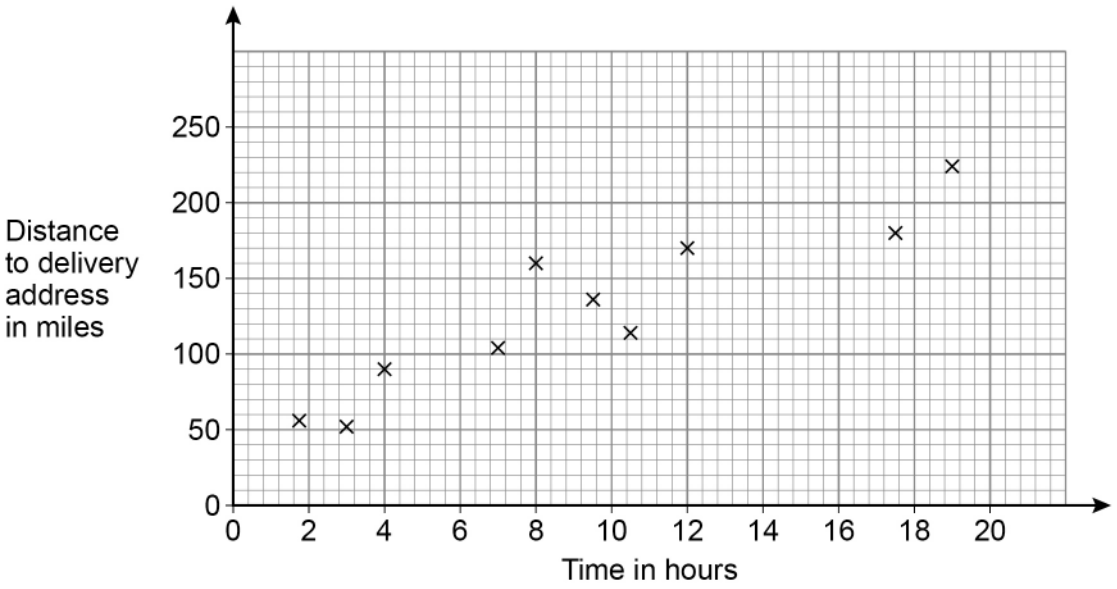
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11 Parcels

Emil owns a company that delivers parcels.

11 (a) Emil records the delivery distance and the time taken for 12 parcels to be delivered.

The scatter diagram shows information for 10 of these parcels.



The table shows the data for the other two parcels.

Time (hours)	Delivery distance (miles)
11	150
16	190

Plot the two extra points and then use the scatter diagram to estimate the delivery distance of a parcel that takes 14 hours to deliver.

You **must** show your working, which should be on the diagram.

State the units of your answer.

[5 marks]

Answer _____

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11 (b) Emil uses this formula to calculate the delivery cost for a parcel.

Cost in pounds = 1.5 × weight in kilograms + 0.04 × delivery distance in miles

A parcel

weighs 4.2 kg

and

has a delivery distance of 85 miles.

Emil says,

"The cost will be **less than** £10"

Is he correct?
You **must** show your working.

[3 marks]

8

END OF QUESTIONS

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Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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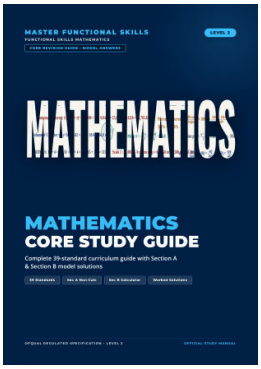
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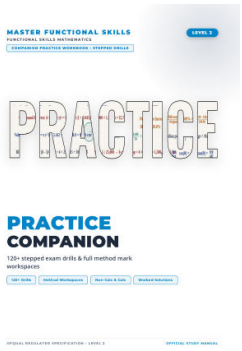


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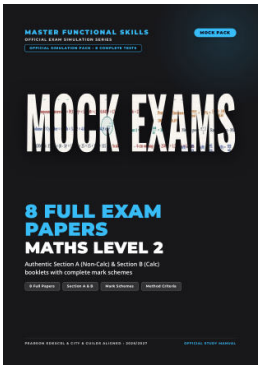


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